

## REPUBLIC OF SOUTH AFRICA

PATENTS ACT, 1978

## APPLICATION FOR A PATENT AND ACKNOWLEDGEMENT OF RECEIPT

## [Section 30 (1)-Regulation 22]

The granting of a patent is hereby requested by the undermentioned applicant on the basis of the present application.

|                          |    |                   |                                  |  |
|--------------------------|----|-------------------|----------------------------------|--|
| Official Application No. |    |                   | Applicant's or Agent's Reference |  |
| 21                       | 01 | <b>2025/10605</b> | 6016LPS                          |  |

|                                                                                                                              |                              |  |  |  |
|------------------------------------------------------------------------------------------------------------------------------|------------------------------|--|--|--|
| 71                                                                                                                           | Full Name(s) of Applicant(s) |  |  |  |
| VISHWAKARMA INSTITUTE OF TECHNOLOGY<br>666, UPPER MARKET RD, UPPER INDIRA NAGAR, BIBWEWADI, PUNE, MAHARASHTRA, 411037, India |                              |  |  |  |

|                                                                                                                                              |                    |  |  |  |
|----------------------------------------------------------------------------------------------------------------------------------------------|--------------------|--|--|--|
| 54                                                                                                                                           | Title of invention |  |  |  |
| <b>A PRIVACY PROTECTION SYSTEM FOR PERSONAL SENSITIVE INFORMATION USING LOCAL AI REDACTION AND VERIFIABLE CREDENTIAL (VC) ACCESS CONTROL</b> |                    |  |  |  |

|                                                                                                         |         |       |
|---------------------------------------------------------------------------------------------------------|---------|-------|
| The applicant claims priority as set out on the accompanying Form P.2. The earliest priority claimed is |         |       |
| COUNTRY:                                                                                                | NUMBER: | DATE: |

|                                                                                   |    |  |
|-----------------------------------------------------------------------------------|----|--|
| <del>This application is for a patent of addition to patent application No.</del> |    |  |
| 21                                                                                | 01 |  |

|                                                                                                        |    |  |
|--------------------------------------------------------------------------------------------------------|----|--|
| <del>This application is a fresh application in terms of section 37 and based on Application No.</del> |    |  |
| 21                                                                                                     | 01 |  |

|                                     |  |  |
|-------------------------------------|--|--|
| This application is accompanied by: |  |  |
|-------------------------------------|--|--|

|   |     |                                                                                                                    |
|---|-----|--------------------------------------------------------------------------------------------------------------------|
| X | 1.  | A single copy of a complete specification of 8 pages.                                                              |
| X | 2.  | Drawings of 1 sheet(s).                                                                                            |
| X | 3.  | Publication particulars and abstract(Form P8)                                                                      |
| X | 4.  | A copy of a figure of the drawing (if any) for the abstract                                                        |
| X | 5.  | Assignment of invention                                                                                            |
|   | 6.  | Certified priority document(s)                                                                                     |
|   | 7.  | Translation(s) of the priority document(s)                                                                         |
|   | 8.  | Assignment of priority rights                                                                                      |
|   | 9.  | A copy of the Form P.2 and the specification of S.A Patent Application (if applicable).                            |
| X | 10. | A declaration and power of attorney on Form P3                                                                     |
| X | 11. | Statement on the use of indigenous Biological Resource, Genetic Resource, Traditional Knowledge or Use on Form P26 |

|                                                                                        |                     |
|----------------------------------------------------------------------------------------|---------------------|
| 74                                                                                     | Address of Service: |
| Wolmarans and Susan Inc.<br>337 Surrey Avenue, Randburg, Gauteng, 2194<br>SOUTH AFRICA |                     |

Dated this 9th day of December 2025

**Digitally signed by : Leon Pierre Susan**

.....

Signature of Applicant(s)

This is returned to the applicant's  
address for service as proof of lodging.

|                               |
|-------------------------------|
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| Official Date Stamp           |
| .....<br>Registrar of Patents |

REPUBLIC OF SOUTH AFRICA

REGISTER OF PATENTS

PATENTS ACT, 1978

|                                                                                                                                              |                                           |                           |                    |                    |      |
|----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|---------------------------|--------------------|--------------------|------|
| Official application No.                                                                                                                     |                                           | Lodging date: Provisional |                    | Acceptance date    |      |
| 21                                                                                                                                           | 01                                        | <b>2025/10605</b>         |                    | 22                 |      |
| International classification                                                                                                                 |                                           | Lodging date: Complete    |                    | Granted date       |      |
| 51                                                                                                                                           | H04L                                      |                           | 23                 | 2025/12/09         |      |
| 71                                                                                                                                           | Full name(s) of applicant(s)/Patentee(s): |                           |                    |                    |      |
| VISHWAKARMA INSTITUTE OF TECHNOLOGY<br>666, UPPER MARKET RD, UPPER INDIRA NAGAR, BIBWEWADI, PUNE, MAHARASHTRA, 411037, India                 |                                           |                           |                    |                    |      |
| 71                                                                                                                                           | Applicant substituted:                    |                           |                    | Date registered    |      |
|                                                                                                                                              |                                           |                           |                    |                    |      |
| 71                                                                                                                                           | Assignee(s):                              |                           |                    | Date registered    |      |
|                                                                                                                                              |                                           |                           |                    |                    |      |
| 72                                                                                                                                           | Full name(s) of inventor(s):              |                           |                    |                    |      |
| KHONDRE, Shruti Pradeep<br>OSWAL, Vidhi<br>JAHANGIR, Umar<br>JAHANGIR, Sidra<br>CHAUDHARI, Prasad B.<br>SHELKE, Priya<br>SURYAWANSHI, Amol   |                                           |                           |                    |                    |      |
| Priority claimed:                                                                                                                            |                                           | Country                   | Number             |                    | Date |
|                                                                                                                                              |                                           |                           |                    |                    |      |
|                                                                                                                                              |                                           |                           |                    |                    |      |
| 54                                                                                                                                           | Title of invention                        |                           |                    |                    |      |
| <b>A PRIVACY PROTECTION SYSTEM FOR PERSONAL SENSITIVE INFORMATION USING LOCAL AI REDACTION AND VERIFIABLE CREDENTIAL (VC) ACCESS CONTROL</b> |                                           |                           |                    |                    |      |
|                                                                                                                                              |                                           |                           |                    |                    |      |
| Address of applicant(s)/patentee(s):                                                                                                         |                                           |                           |                    |                    |      |
| 666, UPPER MARKET RD, UPPER INDIRA NAGAR, BIBWEWADI, PUNE, MAHARASHTRA, 411037<br>INDIA                                                      |                                           |                           |                    |                    |      |
| 74                                                                                                                                           | Address for service                       |                           |                    |                    |      |
| Wolmarans and Susan Inc.<br>337 Surrey Avenue, Randburg, Gauteng, 2194<br>SOUTH AFRICA<br>Reference No. 6016LPS                              |                                           |                           |                    |                    |      |
| 61                                                                                                                                           | Patent of addition No.                    |                           |                    | Date of any change |      |
|                                                                                                                                              |                                           |                           |                    |                    |      |
| Fresh application based on.                                                                                                                  |                                           |                           | Date of any change |                    |      |
|                                                                                                                                              |                                           |                           |                    |                    |      |

REPUBLIC OF SOUTH AFRICA  
PATENTS ACT, 1978  
**COMPLETE SPECIFICATION**  
[Section 30(1) – Regulation 28]

OFFICIAL APPLICATION NO.

|    |    |                   |
|----|----|-------------------|
| 21 | 01 | <b>2025/10605</b> |
|----|----|-------------------|

LODGING DATE

|    |            |
|----|------------|
| 22 | 2025/12/09 |
|----|------------|

INTERNATIONAL CLASSIFICATION

|    |      |
|----|------|
| 51 | H04L |
|----|------|

FULL NAME(S) OF APPLICANT(S)

|    |                                                                                                                              |
|----|------------------------------------------------------------------------------------------------------------------------------|
| 71 | VISHWAKARMA INSTITUTE OF TECHNOLOGY<br>666, UPPER MARKET RD, UPPER INDIRA NAGAR, BIBWEWADI, PUNE, MAHARASHTRA, 411037, India |
|----|------------------------------------------------------------------------------------------------------------------------------|

FULL NAME(S) OF INVENTORS(S)

|    |                                                                                                                                                                 |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 72 | 1. KHONDRE, Shruti Pradeep<br>2. OSWAL, Vidhi<br>3. JAHANGIR, Umar<br>4. JAHANGIR, Sidra<br>5. CHAUDHARI, Prasad B.<br>6. SHELKE, Priya<br>7. SURYAWANSHI, Amol |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|

TITLE OF INVENTION

|    |                                                                                                                                              |
|----|----------------------------------------------------------------------------------------------------------------------------------------------|
| 54 | <b>A PRIVACY PROTECTION SYSTEM FOR PERSONAL SENSITIVE INFORMATION USING LOCAL AI REDACTION AND VERIFIABLE CREDENTIAL (VC) ACCESS CONTROL</b> |
|----|----------------------------------------------------------------------------------------------------------------------------------------------|

REPUBLIC OF SOUTH AFRICA  
PATENTS ACT, 1978  
**PUBLICATION PARTICULARS AND ABSTRACT**  
[Section 32(3)(a) – Regulation 2291)(g) AND 31]

| OFFICIAL APPLICATION NO. |                         | LODGING DATE  | ACCEPTANCE DATE |
|--------------------------|-------------------------|---------------|-----------------|
| 21                       | 01<br><b>2025/10605</b> | 22 2025/12/09 | 47              |

| INTERNATIONAL CLASSIFICATION | NOT FOR PUBLICATION                     |
|------------------------------|-----------------------------------------|
| 51 H04L                      | CLASSIFIED BY: Wolmarans and Susan Inc. |

| FULL NAME(S) OF APPLICANT(S) |                                                                                                                              |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| 71                           | VISHWAKARMA INSTITUTE OF TECHNOLOGY<br>666, UPPER MARKET RD, UPPER INDIRA NAGAR, BIBWEWADI, PUNE, MAHARASHTRA, 411037, India |

| FULL NAME(S) OF INVENTORS(S) |                                                                                                                                                                 |
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| 72                           | 1. KHONDRE, Shruti Pradeep<br>2. OSWAL, Vidhi<br>3. JAHANGIR, Umar<br>4. JAHANGIR, Sidra<br>5. CHAUDHARI, Prasad B.<br>6. SHELKE, Priya<br>7. SURYAWANSHI, Amol |

| EARLIEST PRIORITY CLAIMED |        |      |
|---------------------------|--------|------|
| COUNTRY                   | NUMBER | DATE |
| 33                        | 31     | 32   |

| TITLE OF INVENTION |                                                                                                                                       |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| 54                 | A PRIVACY PROTECTION SYSTEM FOR PERSONAL SENSITIVE INFORMATION USING LOCAL AI REDACTION AND VERIFIABLE CREDENTIAL (VC) ACCESS CONTROL |

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 57 | <p>The present invention is related to a privacy protection system for personal sensitive information using local AI Redaction and Verifiable Credential (VC) Access Control. This invention relates to a privacy-aware camera system capable of performing on-device AI-powered detection and redaction of sensitive information such as human faces, textual data, and vehicle license plates in real time. The system operates entirely at the edge, ensuring that original unprocessed content never leaves the local device without authorization. A mock Verifiable Credential (VC) and Decentralized Identifier (DID) framework is integrated to authenticate and grant controlled access for selective unblurring of redacted regions, enabling context-based privacy enforcement. The architecture consists of five layers: input &amp; capture, AI detection &amp; redaction, access control, selective cloud inference, and logging &amp; compliance. It supports rule-based privacy modes, dynamic configuration, and maintains a complete audit trail through secure logging for regulatory compliance. The system is suitable for privacy-critical applications such as surveillance, document digitization, vehicle monitoring, and secure archival, where strict control over sensitive visual data is required.</p> |
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